

Roll No.....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program : Bachelor of Technology (Biotechnology)

Semester : 1st

Name of the Course : Structural Biology

Course Code : TBT306b

Time : 90 Minutes

Maximum Marks : 50

Note:

- (i) Answer all the questions by choosing *any one of the sub questions*.
- (ii) Each question carries 10 marks

Q1 (10 marks)		CO1
a)	Discuss the significance of the post-NGS era in revolutionizing biological macromolecule research and explain the key steps involved in a typical NGS workflow.	
OR		
b)	Compare traditional Sanger sequencing with Next-Generation Sequencing in terms of principle, throughput, and applications ? (10 marks)	
Q2		CO2
a)	Analyze how non-covalent interactions contribute to stabilizing higher-order protein structures ?	
OR		
b)	Evaluate the role of NGS in understanding genetic variations (mutations, SNPs) and their impact on protein structure. (10 marks)	
Q3		CO3
a)	Explain the importance of motifs and domains in determining protein function.	
OR		
b)	Explain the process of clonal amplification in NGS. Compare emulsion PCR and bridge PCR in terms of mechanism, platforms used, and advantages. (10 marks)	
Q4		CO4
a)	Discuss the concept of "magic pockets" in enzymes and their role in substrate specificity.	
OR		
b)	Discuss how macromolecules can be visualized using computational tools such as RasMol, PyMOL, and Discovery Studio, highlighting their key features and applications in structural biology. (10 marks)	
Q5		CO5
a)	Explain the role of covalent, ionic, hydrophobic, and van der Waals interactions in stabilizing macromolecular structures.	
OR		
b)	What is the Protein Data Bank (PDB)? Explain its role in structural biology research.	

